A3-PS015-TC032

Explainable Home and Last-Mile Connectivity Diagnosis

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Problem Description

A speed test tells you how fast a link can go, but not why it feels broken. We propose a tool that finds the likely cause using real measurements (proposal; not built yet).

Abstract

Most people judge their internet with one speed-test number. That number can miss lost data (packet loss), slow replies (latency), replies that keep changing when the link is busy (jitter) and slow website-name lookups (DNS). We propose a simple desktop tool that guides the user through a few checks, collects measurements, and uses clear rules to say where the problem most likely is.

Problem

- Slow pages or frozen videos can come from very different causes: a **problem at the home router or Wi-Fi**, a **DNS failure**, **packet loss**, or **congestion** (too much traffic on the link).
- To tell them apart we need several measurements, compared with a normal "good" baseline, and the tool must admit when it is not sure.

Objectives

- Measure: can we reach the router, do website names resolve (and how fast), how long replies take, how much data is lost, and how the link behaves when it is busy.
- Explain every answer: show the measurements and the rule used. Say **"Inconclusive"** when the cause cannot be found.
- Create a report with the evidence when the problem cannot be pinned down.

Who it is for: home users, students and first-line support staff. **Goal:** an easy-to-understand diagnosis, to be tested in a controlled lab on four planned fault cases.

Not claimed: proof of a damaged cable, causes inside the ISP's network, or Wi-Fi interference (this needs radio data we do not have).

Background

Basic networking ideas needed to follow the proposal.

- **Home network and default gateway:** devices at home reach the internet through the default gateway, usually the home router. If the router cannot be reached, the problem is probably in the device, Wi-Fi or router.
- **DNS:** DNS turns website names into IP addresses [6], [7]. Names can fail to resolve, or resolve slowly, even when the connection itself works.
- **Latency and round-trip time (RTT):** how long a small test message takes to reach a server and come back [2]. A “ping” (ICMP echo) is a common test message [8].
- **Packet loss:** the share of test messages that get no reply in time [3].
- **Throughput:** how much data is delivered per second. A high top speed does not show delay or lost data, and speed tests must be designed carefully [5].
- **Loaded latency and jitter:** latency measured while the link is busy. Jitter means the delay keeps changing [4]. A busy link often makes both worse.
- **Last-mile access:** the final link to the home: fibre, mobile (cellular), fixed wireless or satellite. Each behaves differently, so we measure a baseline for each connection instead of guessing.
- **Why separate local problems:** Wi-Fi or home-network faults can often be fixed by the user, while others need a call to the provider. Without radio data, the tool will not claim Wi-Fi interference.

Literature & RFC Review

RFC 2330 gives the rules for measuring fairly; our diagnosis rules are our own proposed design.

Existing standards

- **RFC 2330, Framework for IP Performance Metrics [1]:** our main reference. It says measurements should be clearly defined and repeatable, separates one measurement from a set of measurements, says results depend on the type of packet used, and warns that measurement errors must be considered when reading results.
- **Later RFCs built on it:** round-trip delay [2], one-way loss [3] and delay variation [4]. We use similar ideas with ping and DNS tests. Loss measured with ping covers both directions, so it is *not* the same as the one-way loss in [3].
- **RFC 6349, TCP throughput testing [5]:** helps us describe and repeat our load test.

Why it matters for our tool

- **Repeatable tests:** same number of probes, timing, wait time and target every run, and we save these settings.
- **Real evidence:** show a typical value, how much it varies, and the loss percentage, not just one number.
- **Careful reading:** a symptom is a hint, not proof; every report lists its limits.

Our own proposal (not part of any RFC)

The decision rules, confidence levels, “Inconclusive” result and escalation report are our own design. The RFCs explain how to measure, not how to troubleshoot, so our limits are *proposed values that must be tested and tuned first.*

Proposed Solution

A simple desktop tool with a guided, step-by-step test, made for a controlled lab (proposed; not built yet).

Guided testing steps

1. **Start** a test session.
2. **Collect** the network details we can get (connection type, router address, DNS servers). Mark missing items "unavailable".
3. **Check the router (gateway)** can be reached, several times.
4. **Check DNS:** do names resolve, and how fast?
5. **Measure reply time and packet loss** to the router and an outside server.
6. **Compare** normal and busy-link (controlled load) results.
7. **Apply clear rules** to the evidence.
8. **Explain** the likely cause using the measurements.
9. **Create an escalation report** if the cause is not found.

Each check is a separate module with the same output format, so the rules stay easy to read and each module can be tested alone. Repository:
https://github.com/divyapr-27/A3-PS015-TC032.git

Proposed decision rules

Limits below are proposed values that must be tested and tuned.

- **Router unreachable:** repeated router checks fail: likely a problem at the router or home network. We do not say what physically caused it.
- **DNS failure:** name lookups fail while direct IP checks work: likely a DNS problem. We show which DNS server was tested and the limits.
- **Added packet loss:** in the lab we add loss on purpose and compare it with normal loss. We do not guess how the loss happened.
- **Congestion hint:** delay, jitter or loss gets worse when the link is busy: possible congestion (queueing). This is not proof of an ISP fault.
- **Inconclusive:** results disagree, checks cannot run, or no cause stands out: say "Inconclusive" and list the extra tests that would help.

Optimization Proposed

Design comparison only, not test results: speed test alone versus our step-by-step evidence approach.

Item	Speed test only	Our proposed evidence approach
What is measured	Top speed only	Several measurements: router, DNS, reply time, loss and busy-link behaviour
Finding the fault	Weak; a low number does not say where to look	Guided checks go through router, DNS, delay/loss and load
Explanation	No explanation	Clear rules; evidence and rule shown with each answer
Busy-link testing	Normal conditions only	Compares a controlled busy-link test with a normal baseline
What happens next	Generic advice	Evidence-based escalation report; "Inconclusive" when unsure

How this could make diagnosis more useful (still to be tested)

- **Repeatable tests:** same settings every run, as in [1].
- **Baseline:** each result is compared with the same connection when it is not busy, instead of one fixed limit for everyone.
- **Controlled fault injection:** creating known faults in the lab lets us check that the rules give the right answer (planned).
- **Confidence:** the tool says how sure it is, or "Inconclusive", instead of guessing.

Solution Architecture & Design

Visual: proposed design of the diagnostic tool (design only, not built).

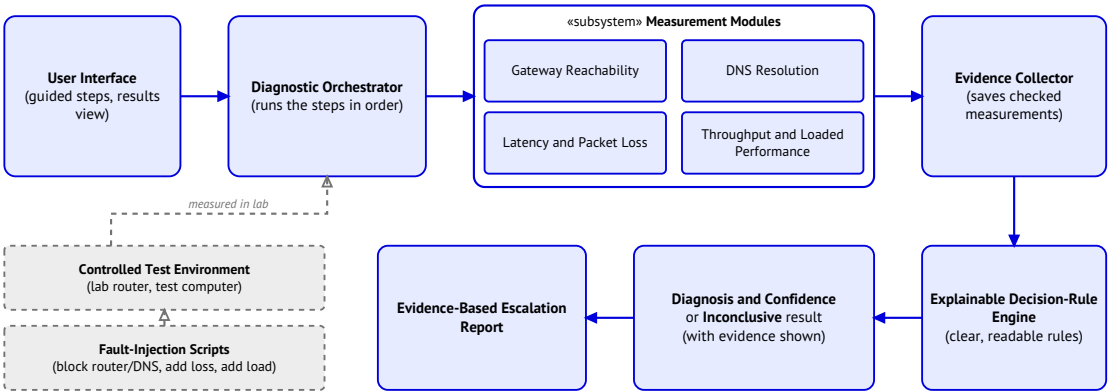


Figure: System Architecture UML Diagram

Data Flow Diagram

Visual: how data moves from the user's start click to a diagnosis, with an Inconclusive path (proposed design).

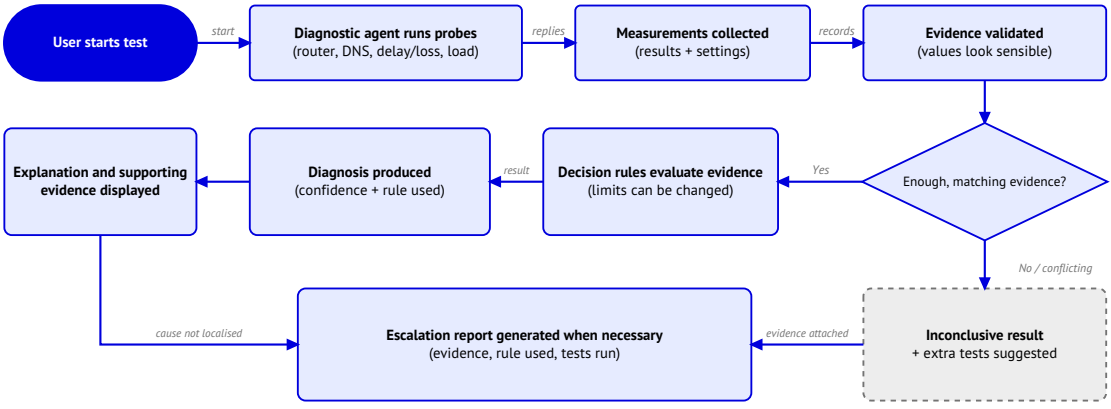


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Proposed plan in relative phases (no dates are claimed as done; real dates to be set by the team).

Phase	Proposed milestone	Intended outcome
Phase 1	**Requirements analysis and measurement design**	Decide what to measure, test settings, evidence format and fault cases [1]
Phase 2	**Desktop diagnostic agent and measurement modules**	Router, DNS, delay/loss and speed/load modules with one output format
Phase 3	**Explainable decision-rule engine**	Easy-to-read rules, confidence levels and an “Inconclusive” result
Phase 4	**Controlled fault-injection scripts**	Lab scripts to block the router or DNS, add packet loss and add load
Phase 5	**Integration and escalation report generation**	Everything works together, including the report
Phase 6	**Evaluation across the four required fault scenarios**	Repeat each fault case against the baseline; report results as seen
Phase 7	**Inconclusive-case analysis**	Study mixed or missing evidence; check the extra tests help
Phase 8	**Final demonstration and documentation**	Demo, user guide and known limits

How we work: step by step. After each phase we review in the lab before starting the next. Phases 6 and 7 report results as they are, including failures.

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Thank You for Your Attention!

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